

ROCHESTER SKYHAVEN, NH, USA

WMO: 726056

Lat: 43.278N

Lon: 70.922W

Elev: 323

StdP: 14.53

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54791

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.4	5.3	-14.5	2.5	3.4	-9.5	3.3	8.5	23.1	20.8	20.7	20.5	6.9	330	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	89.6	72.7	86.3	71.1	83.5	69.5	75.4	84.8	73.6	82.4	72.0	80.0	5.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	121.7	80.0	71.1	115.9	78.2	69.6	110.1	76.7	39.1	84.7	37.4	82.4	35.9	79.7	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	17.2	14.8	-6.4	94.4	4.7	2.1	-9.8	95.9	-12.6	97.2	-15.2	98.4	-18.7	99.9
			-7.4	78.3	4.5	1.4	-10.6	79.4	-13.2	80.2	-15.7	81.0	-18.9	82.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	24.3	26.5	34.2	46.1	56.4	64.9	71.4	69.7	62.3	50.7	39.8	29.9
	DBStd	18.24	10.70	9.58	9.45	8.48	7.77	7.01	5.39	5.43	7.09	7.74	8.38	8.99
	HDD50	3185	798	659	498	174	23	1	0	0	1	87	320	624
	HDD65	6693	1263	1078	955	570	289	86	9	18	134	448	755	1088
	CDD50	2503	1	1	8	56	220	449	663	611	371	107	15	1
	CDD65	536	0	0	0	3	21	84	207	164	54	3	0	0
	CDH74	4868	0	0	10	71	311	795	1824	1373	453	31	0	0
	CDH80	1500	0	0	2	20	94	239	600	415	128	2	0	0
Wind	WSAvg	5.3	6.0	6.3	6.8	6.5	5.4	4.5	3.9	3.8	4.1	4.9	5.5	5.7
Precipitation	PrecAvg	49.1	3.4	3.3	4.7	4.3	4.0	4.1	4.0	3.5	4.0	4.9	4.4	4.3
	PrecMax	66.3	6.8	8.6	15.0	9.6	14.4	12.9	7.5	11.5	10.2	15.4	7.7	7.2
	PrecMin	34.8	0.5	1.1	0.9	0.3	0.8	1.1	0.8	0.2	1.5	0.2	1.9	1.4
	PrecStd	8.1	1.4	1.7	2.9	2.4	3.0	2.6	2.0	2.3	2.5	3.4	1.5	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.0	57.4	70.6	83.2	88.5	91.1	92.6	92.1	89.8	78.3	67.8	57.5
		MCWB	53.1	48.1	55.0	61.0	67.7	73.6	75.2	74.9	73.2	64.7	58.9	52.0
	2%	DB	48.1	49.8	59.3	73.6	81.9	86.2	89.3	88.0	83.6	72.6	61.9	51.3
		MCWB	43.0	43.0	49.1	58.1	64.2	70.9	73.0	72.3	69.8	62.4	55.2	47.2
	5%	DB	43.0	44.8	52.5	66.9	76.6	82.1	86.2	84.4	79.3	67.4	57.3	46.7
		MCWB	38.7	38.7	43.5	54.1	61.3	68.0	71.4	70.8	67.4	60.2	51.6	42.6
	10%	DB	38.4	39.4	47.4	61.8	71.7	78.7	83.2	81.4	75.0	63.2	53.5	42.9
		MCWB	34.3	34.5	40.5	49.5	59.5	65.6	70.3	68.8	65.4	57.3	48.5	39.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.0	51.4	57.7	64.5	70.5	76.1	77.8	77.6	75.0	68.8	61.4	54.4
		MCDB	55.0	54.2	68.8	74.4	84.9	86.6	88.3	88.0	83.0	73.2	64.5	56.6
	2%	WB	44.5	44.0	51.1	60.2	67.0	73.3	75.7	74.9	72.2	64.4	57.0	48.0
		MCDB	46.6	48.4	58.6	71.8	77.1	83.5	85.3	83.5	80.6	69.1	59.9	50.1
	5%	WB	39.0	39.1	45.3	55.8	64.0	70.8	74.0	73.0	70.0	61.6	52.8	43.5
		MCDB	42.4	43.3	50.1	65.5	72.9	78.9	82.8	80.5	76.0	66.2	56.2	46.0
	10%	WB	34.9	35.5	41.1	50.8	60.8	68.2	72.1	71.2	67.7	58.2	49.1	39.2
		MCDB	37.8	39.4	46.9	59.5	69.4	75.0	80.1	78.2	73.2	62.4	53.1	42.3
Mean Daily Temperature Range	5% DB	MDBR	15.2	16.9	17.4	20.1	20.4	19.7	20.1	20.9	20.8	18.3	16.2	14.2
		MCDBR	18.5	21.9	24.7	29.9	29.3	25.2	23.6	24.4	24.4	23.7	21.9	18.9
		MCWBR	16.2	16.4	16.1	16.7	14.4	12.4	10.7	11.2	12.9	14.9	16.5	16.0
	5% WB	MCDBR	18.0	20.8	22.9	27.2	25.5	22.5	20.4	20.5	20.7	20.5	18.4	18.1
		MCWBR	16.5	17.2	16.4	17.5	14.3	12.6	10.4	10.3	12.7	14.3	16.0	16.9
Clear-Sky Solar Irradiance	taub	0.282	0.294	0.318	0.351	0.388	0.434	0.453	0.421	0.369	0.331	0.310	0.289	
	taud	2.462	2.439	2.464	2.426	2.362	2.259	2.213	2.291	2.417	2.529	2.546	2.529	
	Ebn at Noon	273	289	294	290	281	267	260	265	272	272	261	261	
	Edh at Noon	23	28	31	35	38	43	44	39	32	25	21	20	
All-Sky Solar Radiation	RadAvg	540	815	1157	1458	1624	1747	1837	1668	1344	866	578	433	
	RadStd	36	63	73	158	186	153	121	83	80	80	52	47	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	-1.35	N/A	N/A	N/A	N/A
Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148	+56

Nomenclature: See separate page